

# Relativistic Space-Charge Effects with Velocity Binning in OPALX

Semester Project for MSc Physics Student

## Introduction

Particle-in-cell (PIC) simulations are the workhorse for investigating the complex beam dynamics in modern particle accelerators. Central to these simulations is the accurate modeling of self-consistent space-charge forces, typically achieved by solving the POISSON equation,

$$\Delta\Phi \propto \rho,$$

with appropriate boundary conditions in the bunch rest frame i.e. in the electrostatic approximation. The Object-Oriented Parallel Accelerator Library (OPAL) and its performance portable successor, OPALX, provide advanced frameworks for large-scale beam dynamics simulations on modern High-Performance Computing (HPC) architectures.

In many accelerator components, such as during the initial acceleration phase or in presence of strong longitudinal fields, bunches can develop significant energy spreads. In such regimes, the standard approach of assuming a single LORENTZ frame for the entire bunch can lead to and physical inaccuracies (or even numerical difficulties). Relativistic space-charge modelling requires a more nuanced approach where the bunch is partitioned into slices or bins based on energy [1].

The objective of this project is to investigate and validate the energy based binning field solver implementation in OPAL and OPALX.<sup>1</sup>

By employing an energy binning approach, different portions of the particle bunch are treated according to their respective relativistic kinematics. This ensures that relativistic corrections to the self fields are incorporated consistently and computed with physical accuracy.

## OPAL vs. OPALX

OPAL [3] is a production ready tool for particle accelerator simulations. Every time you want to simulate something for this project, OPAL is likely able to do it. OPALX [2] on the other hand is still in heavy development and is supposed to bring GPU compatibility and performance portability to OPAL. These are two different programs. Simulations for this specific project are supposed to be done with OPAL or Python, however, we expect to be able to run these in the new OPALX soon.

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<sup>1</sup>The implementation is currently under development but is expected to become available during the later stages of the project, enabling comparative studies.

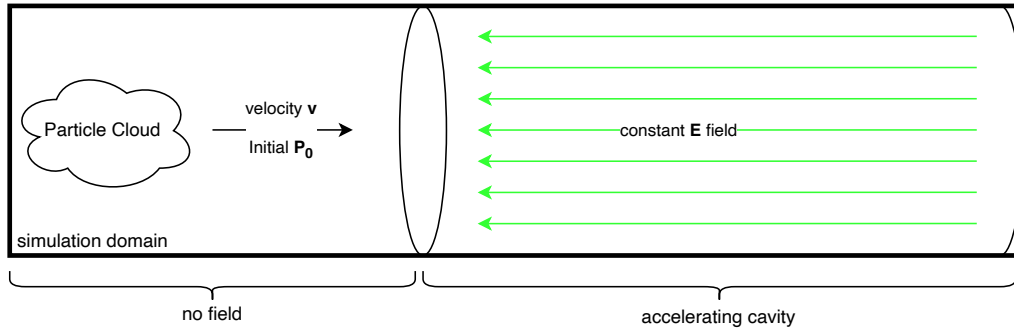


Figure 1: Simple sketch of the setup that is supposed to be studied for this project.

## Project

The project focuses on the characterization of a relativistic electron bunch drifting into a constant longitudinal electric field, see Figure 1. The student will use OPAL to simulate this configuration and validate the results against analytical models and other reference simulations.

The core tasks include:

- **OPAL Simulation and Setup:** Familiarise yourself with the OPAL environment and the `Drift-4-open-bins.in` input [4]. This setup models a GAUSSIAN bunch crossing a drift section into a  $1 \text{ MV m}^{-1}$  longitudinal cavity. Later<sup>2</sup> you might run simulations also in OPALX using the `VELOCITYZ` binning parameter and analyze the evolution of phase space and energy spread.
- **Analytical Modeling:** Derive a (semi-)analytical model for the longitudinal dynamics of a relativistic bunch in a constant  $E_z$  field. Incorporate approximate space-charge effects (e.g., by first using a 1D model) to predict mean position, position spread, energy gain, mean velocity, and energy spread as a function of longitudinal position. This is the main contribution of the project.
- **Reference Python Simulation:** Implement a quasi-1D macroparticle tracker in Python to serve as a baseline. This tool will be used to validate the analytical approximations and provide a direct comparison for the OPAL longitudinal results.
- **Comparative Analysis:** Compare the OPALX binning results<sup>3</sup> with simulations from the original OPAL (without or with different binning). Evaluate the physical accuracy (emittance, bunch length) and numerical stability across different regimes.
- **HPC and Parameter Studies:** Conduct systematic scaling studies on an HPC cluster for the drift test in OPAL (perhaps later in OPALX). Quantify how the simulation performance (runtime, memory) scales with the number of bins, solver grid resolution, and MPI ranks. Explore the transition from the few-bins to the many-bins regime.

## Deliverables

A to-the-point written report and a presentation together with a well documented code and reproducible results (plots etc.) is expected. The documentation must be written such that it is intelligi-

<sup>2</sup>Once the feature is available in OPALX.

<sup>3</sup>Once it is available. Until then, ignore it.

ble to a fellow-student. All items of the project must be pushed to a shared GitHub repository.

## Contacts

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## Literature

- [1] G. Fubiani et al. "Space charge modeling of dense electron beams with large energy spreads". In: Phys. Rev. ST Accel. Beams 9 (6 June 2006), p. 064402. doi: 10.1103/PhysRevSTAB.9.064402. <https://link.aps.org/doi/10.1103/PhysRevSTAB.9.064402>
- [2] OPALX-project at PSI, OPALX Repository, <https://github.com/OPALX-project/OPALX>, (last viewed 03/02/2026).
- [3] OPALX-project at PSI, OPAL Repository, <https://github.com/OPALX-project/OPAL>, (last viewed 03/02/2026).
- [4] OPALX-project at PSI, Drift-4-open-bins.in, <https://github.com/OPALX-project/regression-tests-x/blob/72deeea4e82320428d7a23687d7605dbb57dd588/RegressionTests/Drift-4-open-bins/Drift-4-open-bins.in>, (last viewed 03/02/2026).